H7: Electrodynamics

7.1 electromotive force

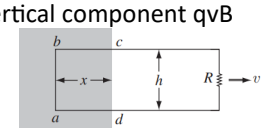
7.1.1 Ohm's law

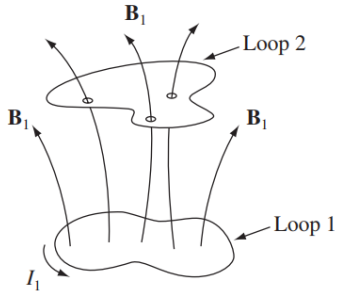
conductivity σ	= ratio between current density **J** and force per unit charge **f**: $\mathbf{J} = \sigma \mathbf{f}.$
resistivity ρ	= 1/σ
Ohm's law	The force **f** is due to the electromagnetic force, thus we can write: $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$ where **vxB** can be ignored, since the velocity is sufficiently small: $\mathbf{J} = \sigma \mathbf{E}.$
Ohm's law in terms of V	the current flowing from one electrode to another is given by: $V = IR.$ > R is the *resistance* with unit ohm Ω
Joule heating law	$P = VI = I^2 R.$

7.1.2 electromotive force

current through a circuit	The charges don't move very fast through the circuit (not speed of light, way slower) > though charges move instantly when connecting a battery > current isn't induced due to charges moving each other nl: its induced due to the electric field > works instantly
electromotive force	2 forces driving charges through a circuit: - the source: q$\mathbf{f_s}$ - electrostatic force: q**E** > smooths out flow + influence of charges we thus have: $\mathbf{f} = \mathbf{f}_s + \mathbf{E}.$ now the emf is defined as: $\mathcal{E} \equiv \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}.$ >> work done per unit charge
emf of a battery	for an ideal source of emf, the net force on the charges is zero > **E** = -$\mathbf{f_s}$, thus the potential difference is: $V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}$ however, batteries have an intrinsic resistance r, thus we have V=ε-Ir

7.1.3 motional emf

motional emf	= emf which arises due to a change in magnetic field
primitive motional emf	We have a uniform magnetic field **B** and resistor R through a loop with height h > loop is pulled with speed v through the field > charges in segment ab experience a magnetic force with vertical component qvB > the emf is: $\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh,$ 
work in a motional emf	the magnetic field can't do work > what force is doing the work in the motional emf?? (figures see p308) >> the person pulling the loop nl: in the loop, the charges have a vertical velocity **u** and horizontal velocity **v** (the pulling) > the magnetic force on the charges has a component quB to the left > to counteract this, the person pull on the wire: f_{pull} = uB to the right then, the particle is moving in the direction of the resultant velocity **w** > this charge goes a distance hlcosθ > work done per unit charge is then: $\int \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} = (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta = vBh = \mathcal{E}$

magnetic flux	= amount of field lines through a surface: $\Phi \equiv \int \mathbf{B} \cdot d\mathbf{a}.$
flux rule	for an emf we have: $\mathcal{E} = -\frac{d\Phi}{dt}.$
vb: eddy currents	take a pendulum with a metallic disk > swing it through a magnetic field > disc will fully stop, due to currents induced in the disc 
7.2 electromagnetic induction	
7.2.1 Faraday's law	
Faraday's law	a changing magnetic field induces an electric field > now we have, for a changing magnetic field: $\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}.$ then, E is related to B by: $\oint \mathbf{E} \cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}.$ and in differential form: $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$
Lenz's law	= nature abhors a change in flux
7.2.2 the induced electric field	
Biot-Savart for E	If we have E in a pure Faraday field, ie E purely due to a changing B field > we then have the laws: $\nabla \cdot \mathbf{E} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$ which resemble the laws of magnetostatics: $\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$ >> we can thus, in this situation, find a law: $\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B}/\partial t) \times \hat{\boldsymbol{\imath}}}{\imath^2} d\tau = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\boldsymbol{\imath}}}{\imath^2} d\tau,$
quasistatic situations	For changing magnetic fields don't adhere to the laws of magnetostatics > however, they hold for fields that don't fluctuate very rapidly or form point far from source > we can still use them as approximations in quasistatic situations
7.2.3 inductance	
Neumann formula	Suppose we have two loops at rest, with a current I_1 going through the first > produces a magnetic field B_1, via Biot-Savart: $\mathbf{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\mathbf{l}_1 \times \hat{\boldsymbol{\imath}}}{\imath^2},$  the field is thus dependent on I_1 > the flux through the loop also depends on I_1: $\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2. \quad \Rightarrow \quad \Phi_2 = M_{21} I_1,$ with M_{21} the mutual inductance of the two loop Now we can solve Φ_2 via: $\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 = \int (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 = \oint \mathbf{A}_1 \cdot d\mathbf{l}_2.$ with $\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\mathbf{l}_1}{\imath},$ hence: $\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\mathbf{l}_1}{\imath} \right) \cdot d\mathbf{l}_2.$ thus we find: $M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{\imath}.$

mutual inductance	If we change the two loop, we should find the same M > we have: $M_{21} = M_{12}.$ thus: the flux through 2 when we run a current I around 1 is identical to the flux through 1 when we send the same current I around 2
self-inductance	A current also induces an flux in the source loop: $\Phi = LI.$ a changing current induces an emf that works against the induced current > thus we have a *back-emf* given by: $\mathcal{E} = -L\frac{dI}{dt}.$
7.2.4 energy in magnetic fields	
energy of a current flowing in a circuit	The amount of work we have to deliver is the work we must do against the back-emf ie: we must work against the induced emf of the circuit > we have: $\frac{dW}{dt} = -\mathcal{E}I = LI\frac{dI}{dt}.$ if we begin at zero and build up to a final current I, we have: $W = \frac{1}{2}LI^2.$
W in terms of B	We have the flux: $\Phi = \int \mathbf{B}\cdot d\mathbf{a} = \int (\nabla \times \mathbf{A})\cdot d\mathbf{a} = \oint \mathbf{A}\cdot d\mathbf{l},$ or thus: $LI = \oint \mathbf{A}\cdot d\mathbf{l},$ we can express W as: $W = \frac{1}{2}I\oint \mathbf{A}\cdot d\mathbf{l} = \frac{1}{2}\oint (\mathbf{A}\cdot \mathbf{I})\, dl.$ which we can generalise to volume currents: $W = \frac{1}{2}\int_{\mathcal{V}} (\mathbf{A}\cdot\mathbf{J})\, d\tau.$ Now, use Ampères law: $W = \frac{1}{2\mu_0}\int \mathbf{A}\cdot(\nabla\times\mathbf{B})\, d\tau.$ which simplifies to: $W = \frac{1}{2\mu_0}\left[\int B^2\, d\tau - \int \nabla\cdot(\mathbf{A}\times\mathbf{B})\, d\tau\right]$ $= \frac{1}{2\mu_0}\left[\int_{\mathcal{V}} B^2\, d\tau - \oint_{\mathcal{S}} (\mathbf{A}\times\mathbf{B})\cdot d\mathbf{a}\right],$ with S the surface bounding V but we can integrate over any volume V > just integrate over all space, thus the surface integral goes to zero and we get: $W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2\, d\tau.$

7.3 Maxwell's equations

7.3.1 electrodynamics before Maxwell

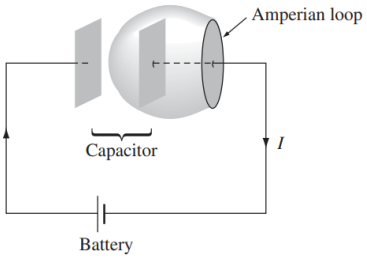
problem with Ampère's law

for the equation: $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$
if we take the divergence, it should be zero:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J});$$

however, the right-hand side generally isn't zero

Also in integral form: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$



now consider a capacitor and battery
> we can choose any surface to integrate over
> chose one that goes through the capacitor
> $I_{enc} = 0$ which we cant find on the lefthand side

7.3.2 how Maxwell fixed Ampère's law

Maxwell's equation

We can make the righthand side of the equation zero by looking at what ∇J is:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t}(\epsilon_0 \nabla \cdot \mathbf{E}) = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right).$$

thus, we can simply insert this term to negate the ∇J:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

>> changing electric fields induce magnetic fields

Now, we can also write this in differential form:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \int \left(\frac{\partial \mathbf{E}}{\partial t}\right) \cdot d\mathbf{a}.$$

7.3.3 Maxwell's equations

Maxwell's equations

$$\begin{aligned}
&\text{(i)} && \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho && \text{(Gauss's law),}\\
&\text{(ii)} && \nabla \cdot \mathbf{B} = 0 && \text{(no name),}\\
&\text{(iii)} && \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law),}\\
&\text{(iv)} && \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} && \text{(Ampère's law with Maxwell's correction).}
\end{aligned}$$

Lorentz force law

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

continuity equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t},$$

7.3.4 magnetic charge

Maxwell's eq in free space

In free space ρ and J vanish, to give us:

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$
$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

7.3.5 maxwell's equations in matter	
polarization current	consider a polarized material with charge density σ_b=P at one end and -σ_b at the other > if P increases a bit, the charge on each end increases accordingly > this gives a net current: $$dI = \frac{\partial \sigma_b}{\partial t} da_\perp = \frac{\partial P}{\partial t} da_\perp.$$ therefore the current density is: $$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}.$$ this polarization current has nothing to do with the bound current density > this is consistent with the continuity eq: $$\nabla \cdot \mathbf{J}_p = \nabla \cdot \frac{\partial \mathbf{P}}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{P}) = -\frac{\partial \rho_b}{\partial t}.$$
Gauss's law in matter	the total charge density can be separated into two parts: $$\rho = \rho_f + \rho_b = \rho_f - \nabla \cdot \mathbf{P},$$ and the current density into three parts: $$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p = \mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}.$$ Gauss's law can now be written as: $$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}(\rho_f - \nabla \cdot \mathbf{P}),$$ or thus: $$\nabla \cdot \mathbf{D} = \rho_f,$$
Ampère's law in matter	Ampère's law becomes: $$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J}_f + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \right) + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$ or thus: $$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t},$$ where we can define the displacement current: $$\mathbf{J}_d \equiv \frac{\partial \mathbf{D}}{\partial t}.$$
Maxwell's eqn in matter	we can now write: (i) $\nabla \cdot \mathbf{D} = \rho_f$, (iii) $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, (ii) $\nabla \cdot \mathbf{B} = 0$, (iv) $\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$.